

MeetingRecorder — Feature Sheet

XIAO ESP32-S3 voice recorder with cloud transcription, live summaries, and an integrated web UI.

Audio capture

- On-board PDM mic (XIAO ESP32-S3 Sense, GPIO 41/42)
- 16 kHz, 16-bit mono PCM — ~32 KB/sec
- 15-second WAV chunks streamed to microSD
- 2× software gain on captured samples
- Real-time capture on a dedicated Core 0 task

WiFi & networking

- Dual-mode WiFi — home network (STA) + always-on hotspot (AP)
- AP survives STA failures, retries, and credential changes
- Hotspot SSID and password configurable from the UI
- STA auto-reconnect with progressive back-off
- NTP sync (pool.ntp.org, Google, Cloudflare) — IST default
- Browser-time fallback when NTP is unavailable

Cloud transcription & summarisation

- ElevenLabs Scribe v1 for speech-to-text
- OpenAI GPT-4o-mini for rolling and final summaries
- 3-attempt retry with WiFi-bounce between tries
- Multipart streaming upload — no full-file RAM buffering
- Rolling summary every chunk once ≥20 words spoken
- Final summary on stop, with fallback if the API fails

Web UI (Dashboard)

- Dark-themed dashboard on port 80 (Inter + JetBrains Mono)
- Live polling every 2.5 s; pauses when the tab is hidden
- Mobile-responsive layout with bottom nav; desktop sidebar
- Transparent mobile status bar — no black strip on iOS/Android
- techiesms branding pill (clickable to YouTube)
- Animated status pill: Idle · Recording · Processing · Done
- Live stats: chunks, words, duration, status icon
- Network and device cards: IP, RSSI, RAM, uptime

Transcript & summary views

- Live rolling transcript updates per chunk
- Final transcript snapshot preserved post-meeting
- Truncation banner above ~5,500 chars
- Markdown rendering: headings, lists, bold/italic
- Copy buttons with HTTP fallback (execCommand)
- Download buttons save transcript and summary as .txt

History & past meetings

- Auto-archived to /meeting_<timestamp>/ on the SD card
- History tab lists newest-first with previews
- Show more / show less inline expansion
- Per-meeting download summary
- Delete (with confirmation) wipes WAVs + transcripts + summaries
- “Ask AI” scoped to any past meeting’s summary

AI Q&A chat

- Ask anything about a meeting once final summary exists
- Switch between current and past meeting contexts
- 6-turn conversation memory for follow-ups
- Suggested prompts (decisions, action items, summary...)
- Themed error rendering via [ERR] sentinel

Configuration

- /setup page for WiFi creds, API keys, AP rename
- /config.json on SD for persistent storage
- No re-flashing required to change credentials
- AP auto-restarts after rename — response delivered first

Concurrency & reliability

- Three pinned FreeRTOS tasks: audio (Core 0), web + processing (Core 1)
- chunkMutex guards chunk handoff between tasks
- stateMutex guards transcript/summary Strings (no realloc races)
- Snapshot pattern: short locks, slow I/O outside the critical section
- Web UI stays responsive during slow STT/GPT calls
- Heap-aware live transcript trim (12 KB max in RAM)

Physical controls

- Hardware button on D1 toggles meeting start/stop
- Built-in LED status (ON = idle, OFF = recording)
- No extra hardware beyond the Sense board + SD card

Developer experience

- Zero-install bundled Shubh’Json JSON parser
- Clean folder structure — src/ modules by concern
- Full docs in docs/: README, ARCHITECTURE, FILES, LICENSE
- Compile-verified on Arduino IDE 2.x + ESP32 core 3.0.0
- 6 unused JSON features disabled at compile time

Resource footprint

Resource	Used	Available
Flash	1.23 MB	3.34 MB (36%)
RAM	40 KB	320 KB (12%)